

Hossam Aly

Dusty Warps in the Local Frame: Instability and Fast Clumping

Collaborators:

Callum Fairbairn (Princeton)

Rebecca Nealon (Monash)

Sijme-Jan Paardekooper (TU Delft)

Gordon Ogilvie (DAMTP)



European Research Council
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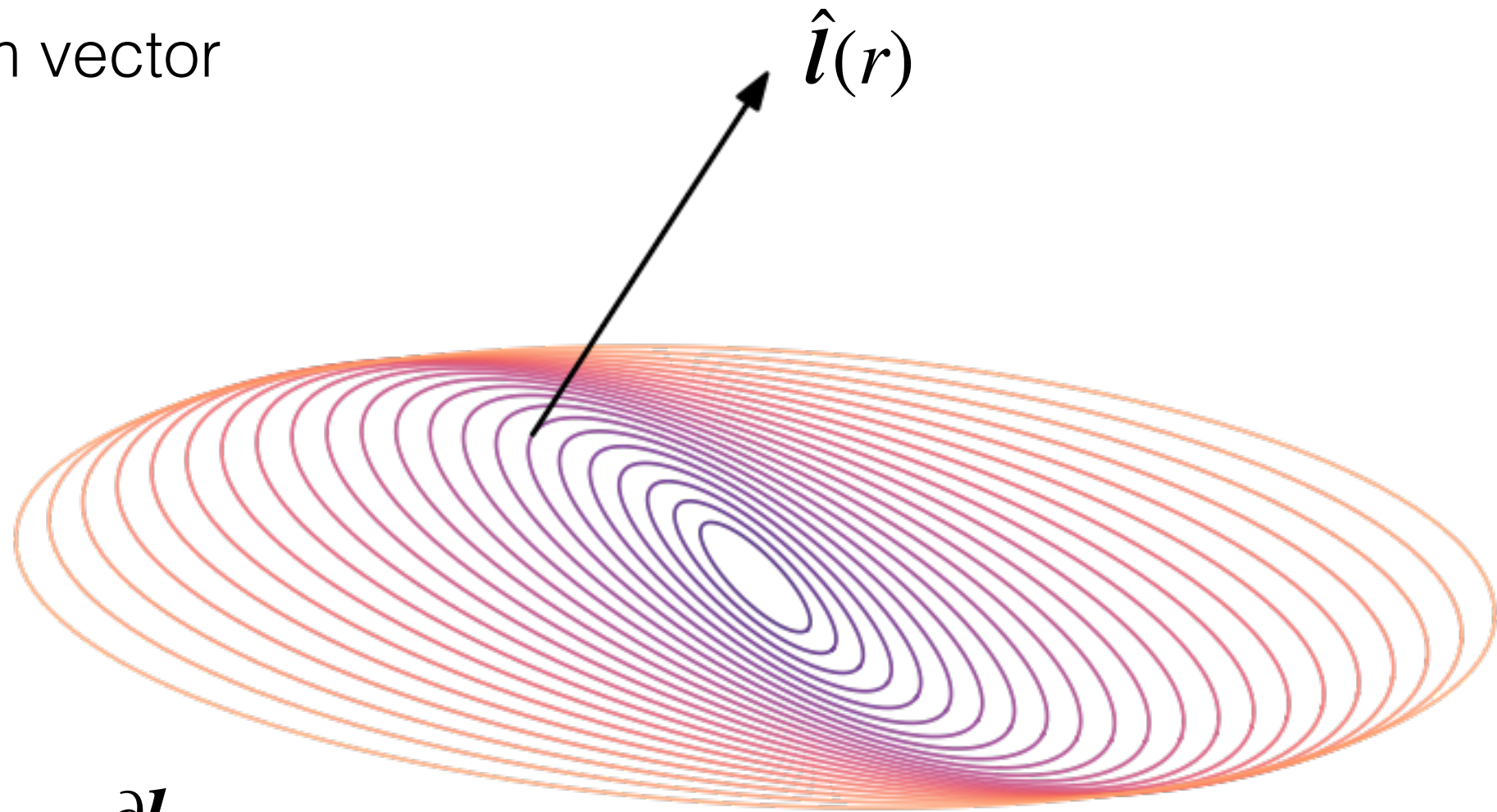
What's a warped disc?

Unit angular momentum vector

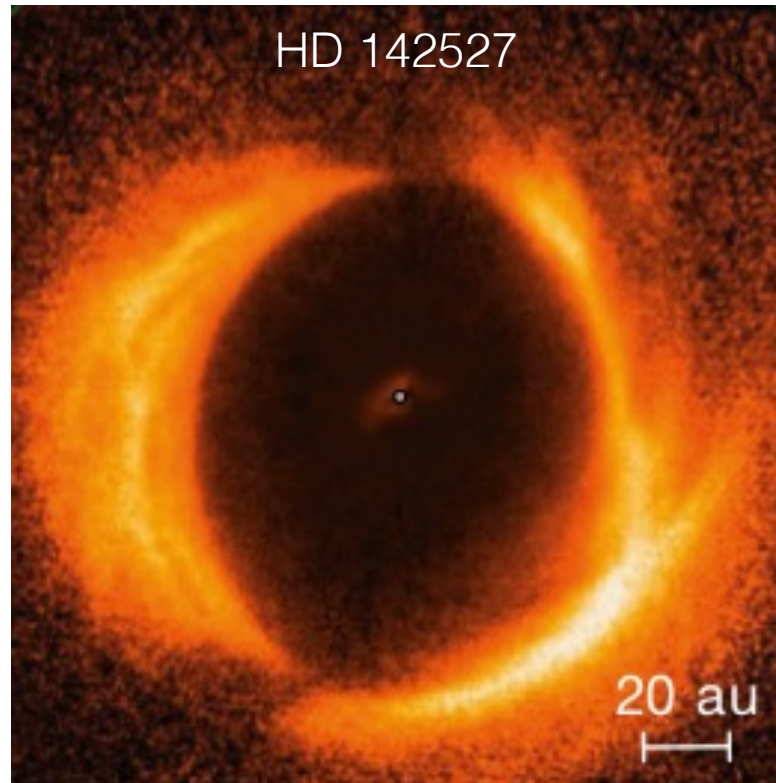
$$\text{Tilt} = \beta = \cos^{-1}(l_z)$$

$$\text{Twist} = \gamma = \tan^{-1}\left(\frac{l_y}{l_x}\right)$$

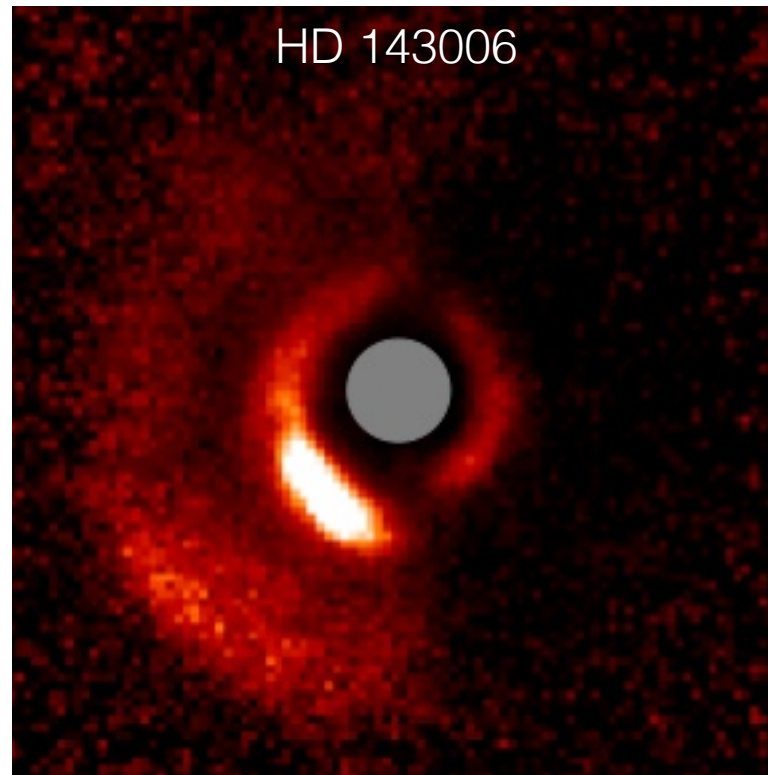
$$\text{Warp amplitude} = \psi = \frac{\partial l}{\partial \ln(r)}$$



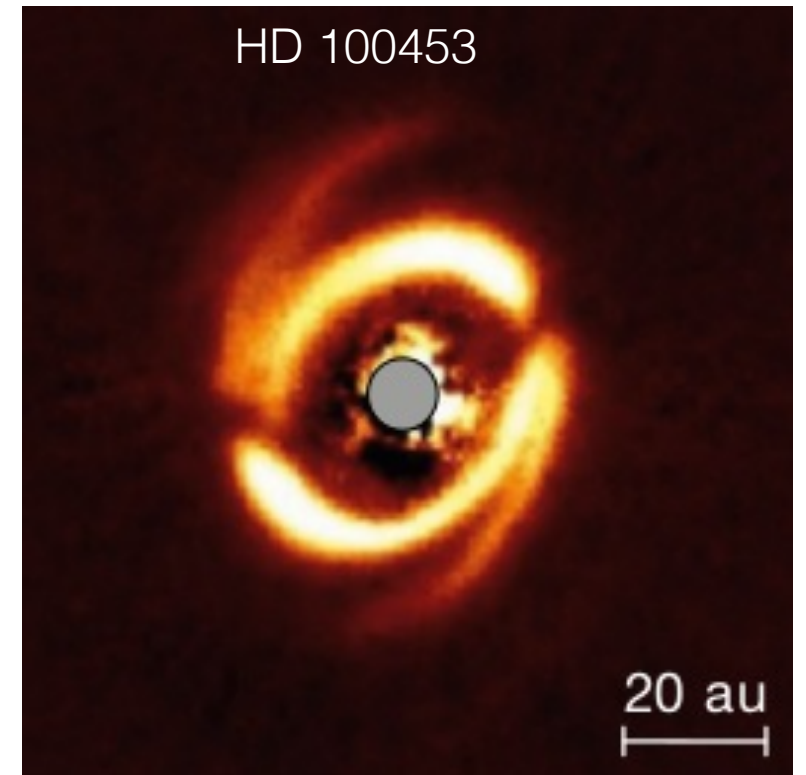
Observational interest in warps: shadows in NIR



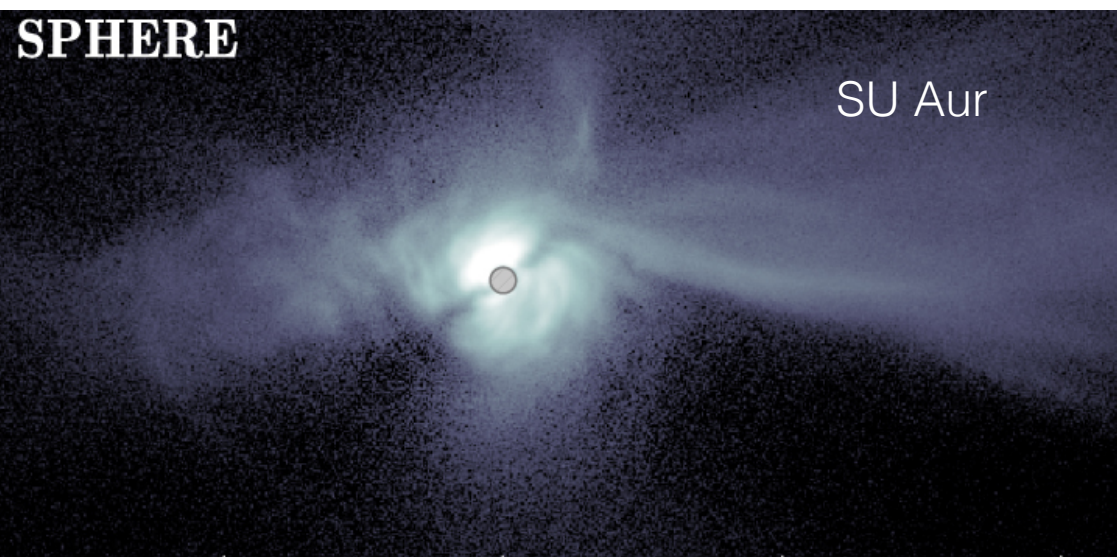
Avenhaus+ 17



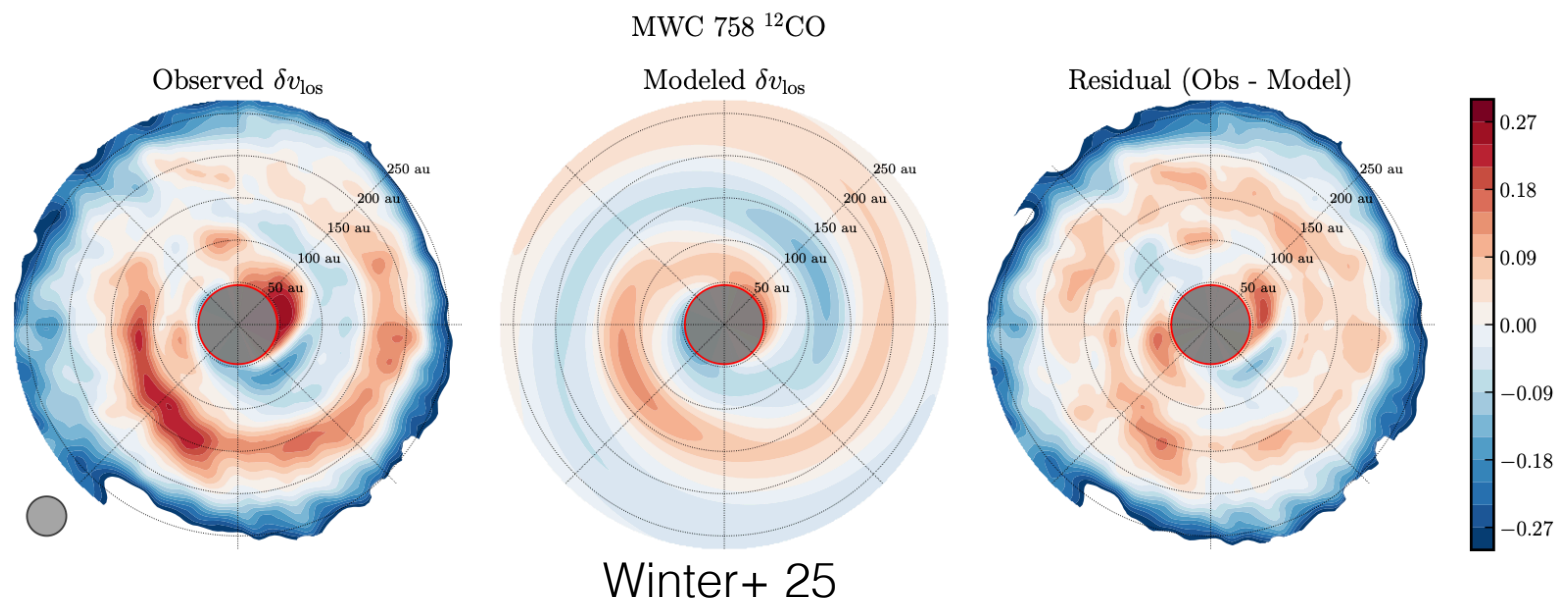
Benisty+ 18



Benisty+ 17



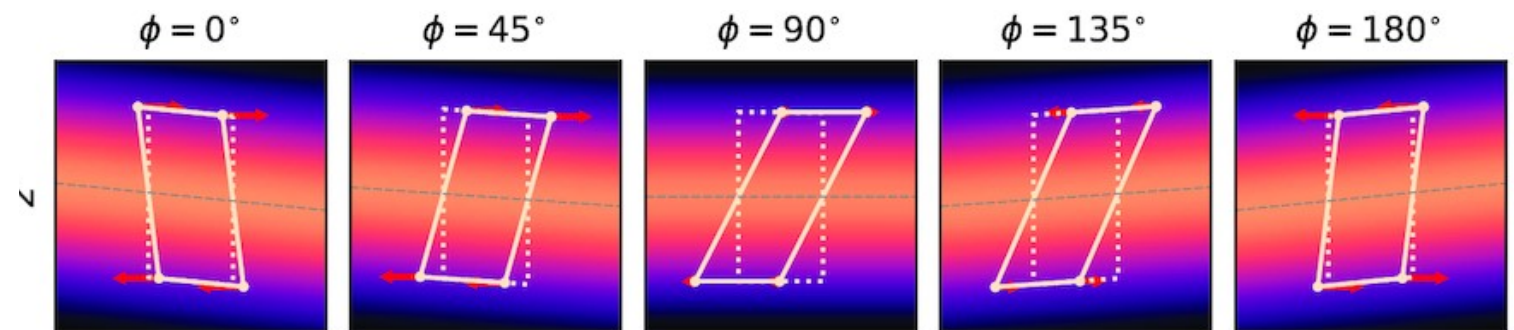
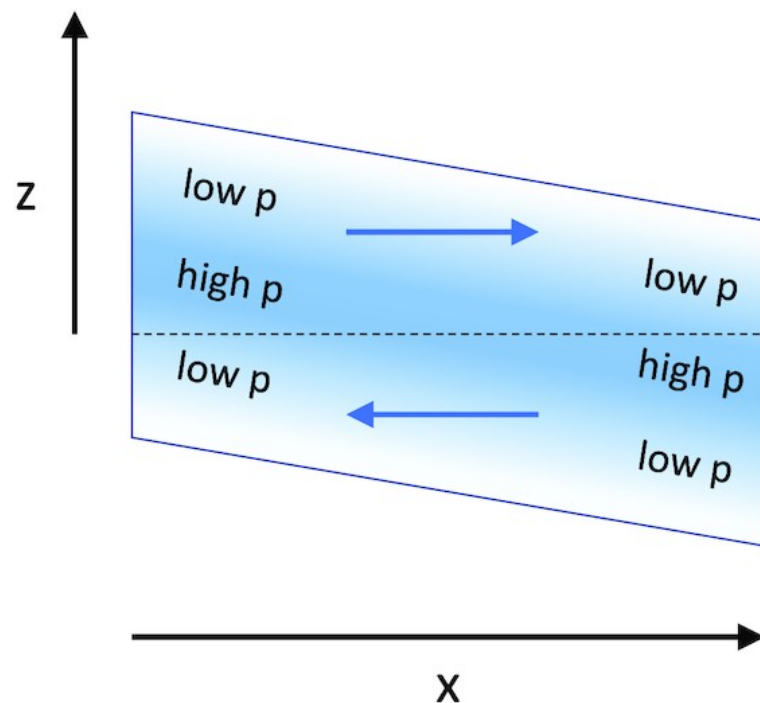
Ginski+ 23



Theory of gas warps

Sloshing (horizontal flows)

Lodato & Pringle 07
Ogilvie & Latter 13a,
Dullemond+ 22

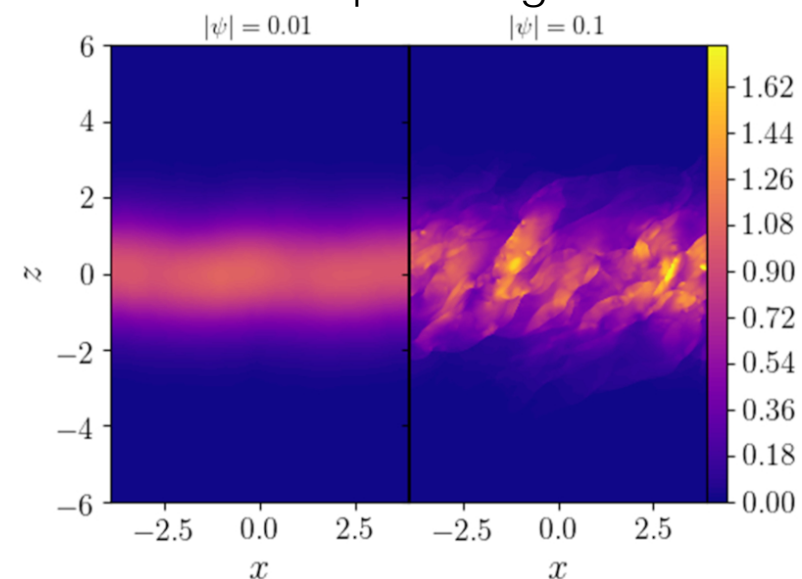


Parametric Instability:

Gammie, Goodman, & Ogilvie 2000

Ogilvie & Latter 13b

Paardekooper & Ogilvie 19



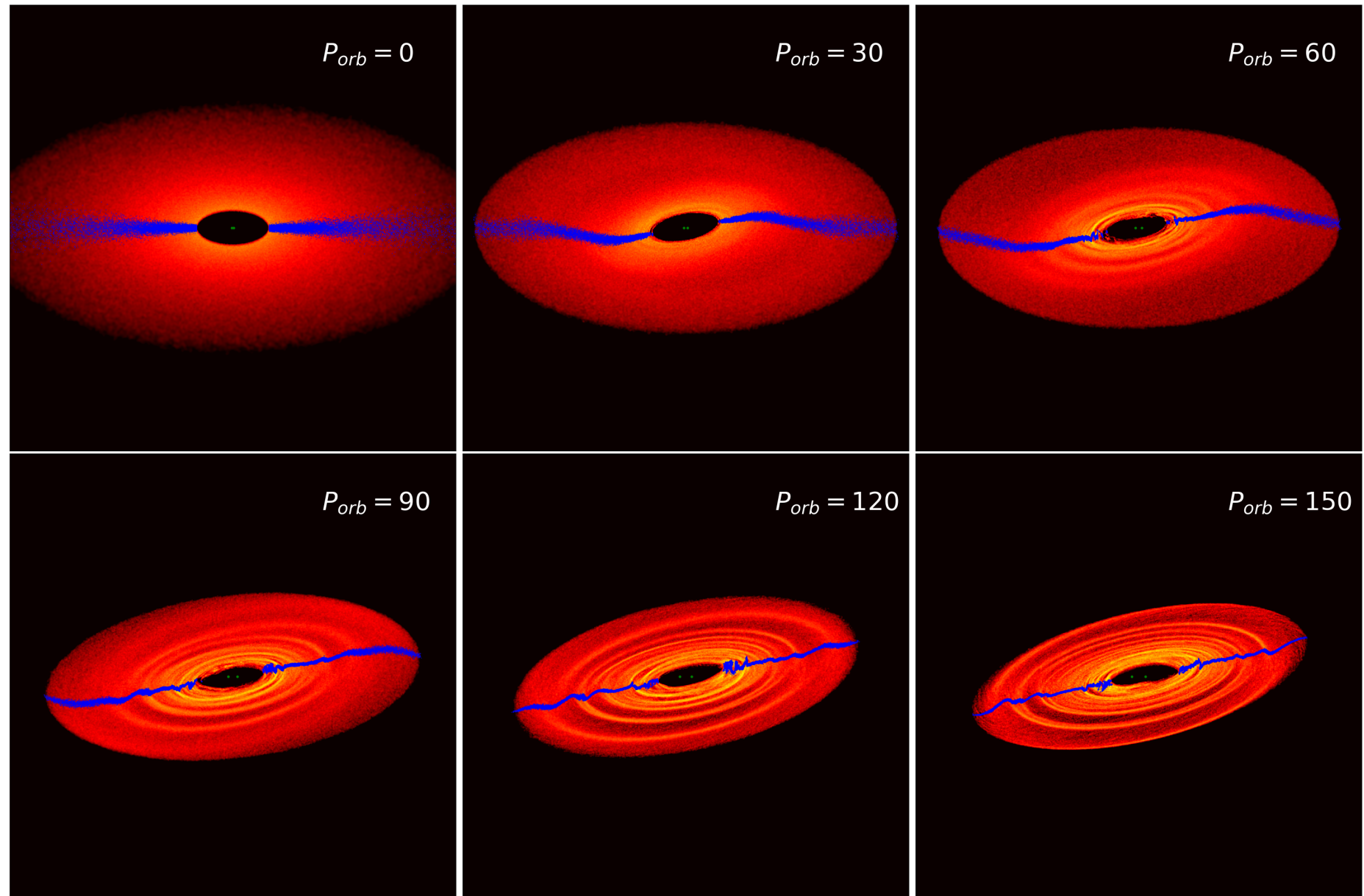
Motivation: Global Warps (WInDI)

Messy simulations!

Questions:

-Global manifestation of a local instability?

-Plays a role in planet formation?



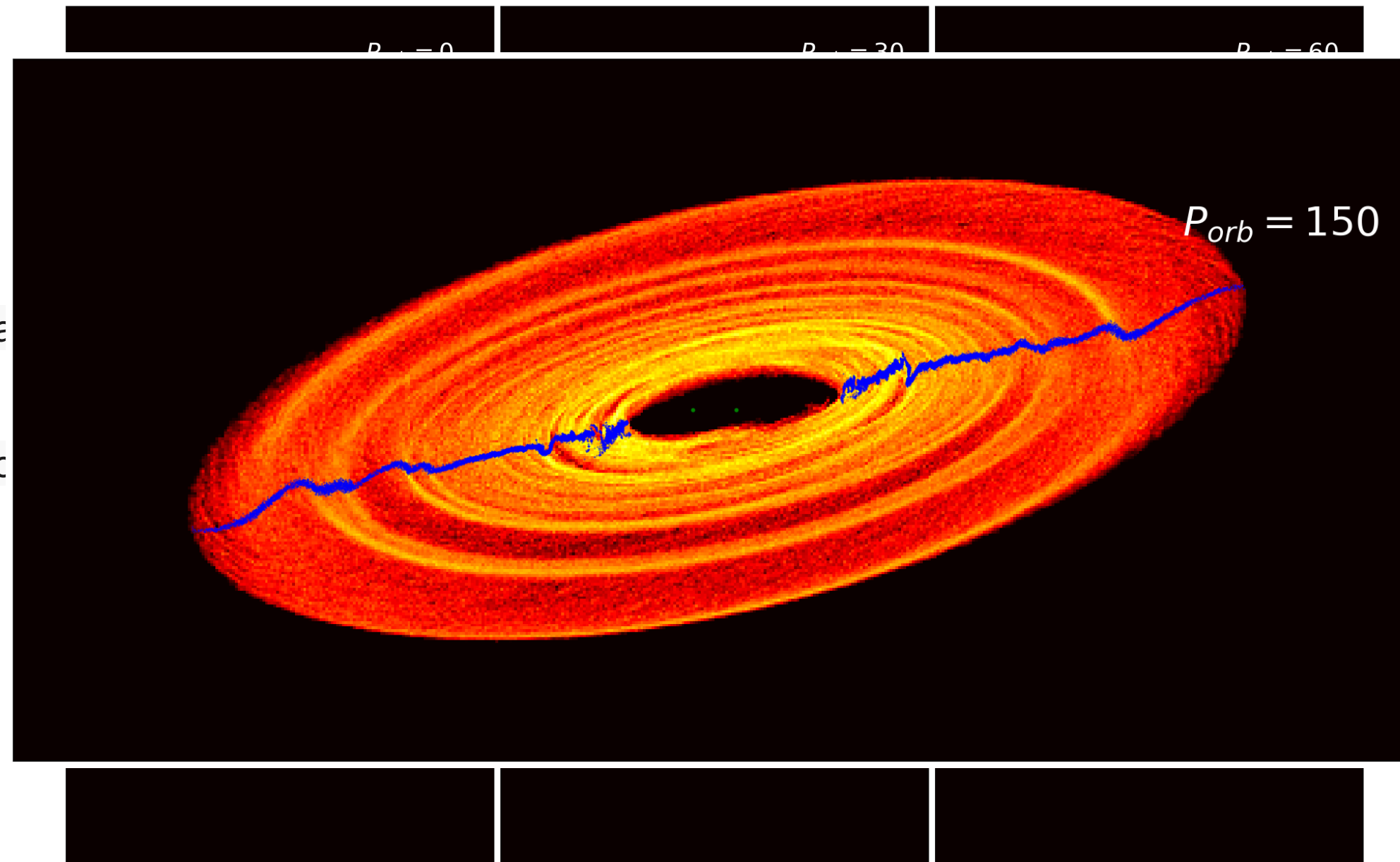
Aly+2024

Motivation: Global Warps (WInDI)

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Questions:

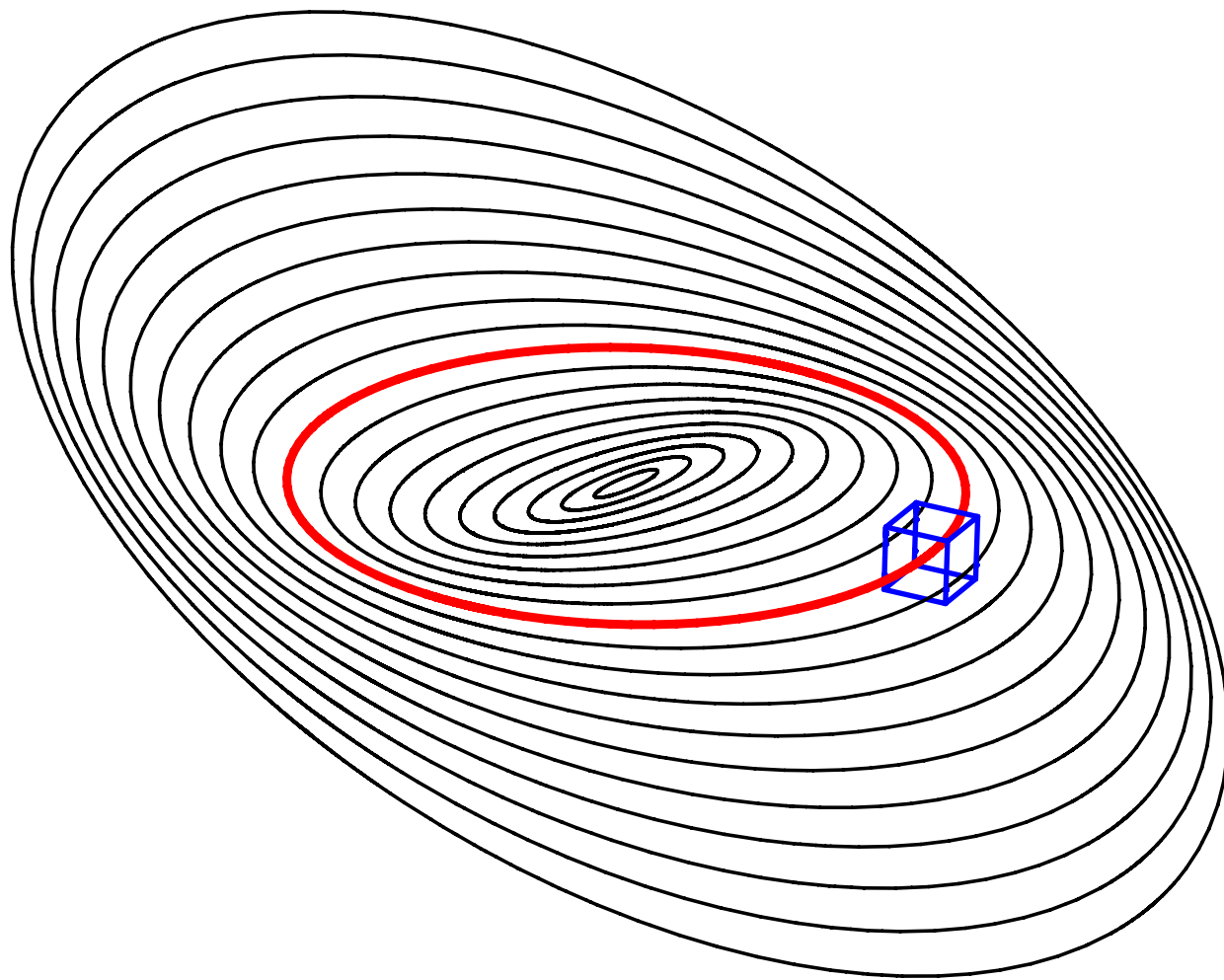
- Global manifestation of a local instability?
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Proper way to model local warps

Warped shearing box (Ogilvie & Latter 13a)



$$t' = t,$$

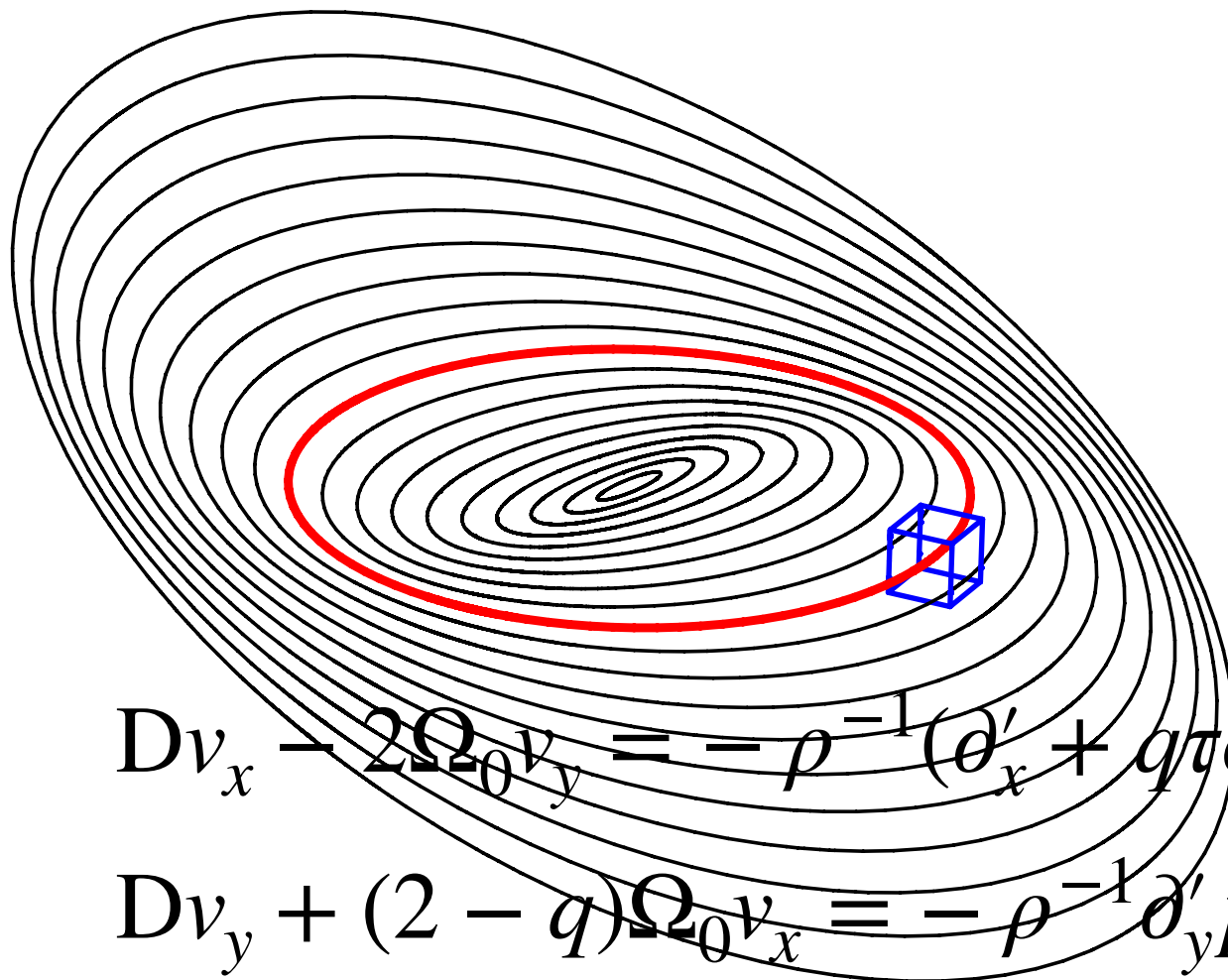
$$x' = x,$$

$$y' = y + q\tau x,$$

$$z' = z + |\psi| \cos \tau x,$$

Proper way to model local warps

Warped shearing box (Ogilvie & Latter 13b)



$$\begin{aligned} t' &= t, \\ x' &= x, \\ y' &= y + q\tau x, \\ z' &= z + |\psi| \cos \tau x, \end{aligned}$$

$$Dv_x - 2\Omega_0 v_y = -\rho^{-1}(\partial'_x + q\tau\partial'_y + |\psi| \cos \tau \partial'_z)p,$$

$$Dv_y + (2 - q)\Omega_0 v_x = -\rho^{-1}\partial'_y p,$$

$$Dv_z + |\psi| \Omega_0 \sin \tau v_x = -\Omega_0^2 z' - \rho^{-1}\partial'_z p,$$

$$D\rho = -\rho[(\partial'_x + q\tau\partial'_y + |\psi| \cos \tau \partial'_z)v_x + \partial'_y v_y + \partial'_z v_z],$$

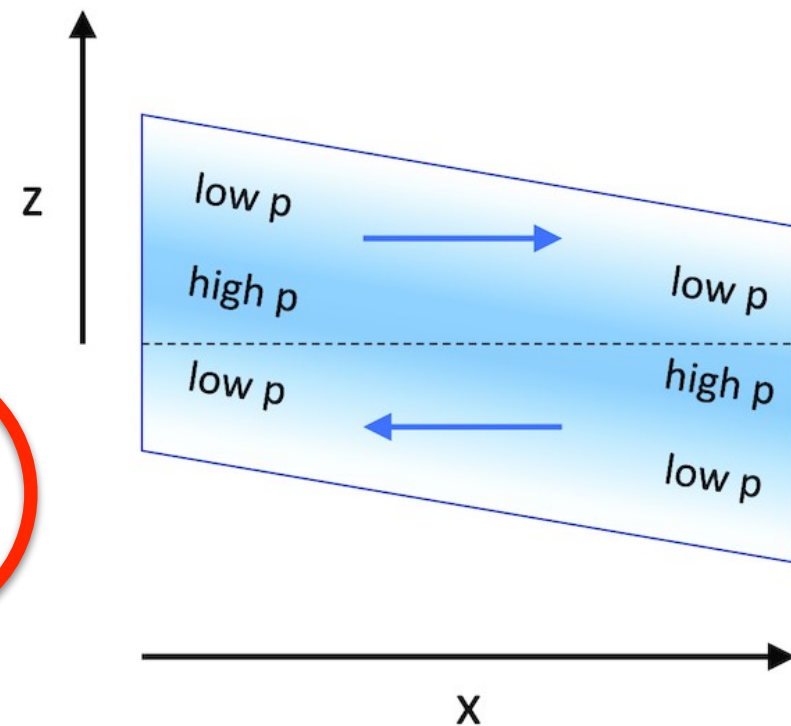
$$D = \partial'_t + v_x \partial'_x + (v_y + q\tau v_x) \partial'_y + (v_z + v_x |\psi| \cos \tau) \partial'_z.$$

Lazy way to model local warps

Usual shearing box, but with forced sloshing

$$\frac{\partial \mathbf{u}_g}{\partial t} + \mathbf{u}_g \cdot \nabla \mathbf{u}_g + 2\mathbf{\Omega} \times \mathbf{u}_g = 2\Omega S x \hat{\mathbf{x}} - \frac{\nabla P}{\rho_g} + \frac{\rho_d}{\rho_g} \frac{\mathbf{u}_d - \mathbf{u}_g}{t_s} + F_w \hat{\mathbf{x}}$$

$$\frac{\partial \mathbf{u}_d}{\partial t} + \mathbf{u}_d \cdot \nabla \mathbf{u}_d + 2\mathbf{\Omega} \times \mathbf{u}_d = 2\Omega S x \hat{\mathbf{x}} - \frac{\mathbf{u}_d - \mathbf{u}_g}{t_s}$$



$$F_w = \text{Re}(\psi \Omega^2 z e^{i\Omega t})$$

Lazy way to model local warps

We look for laminar, constant density solutions

We drop the pressure gradient and assume horizontal homogeneity

$$\frac{du_{b,g}}{dt} - 2\Omega v_{b,g} = \frac{f}{t_s}(u_{b,d} - u_{b,g}) + \psi\Omega e^{i\Omega t}$$

$$\frac{dv_{b,g}}{dt} + \frac{\kappa^2}{2\Omega}u_{b,g} = \frac{f}{t_s}(v_{b,d} - v_{b,g})$$

$$\frac{du_{b,d}}{dt} - 2\Omega v_{b,d} = -\frac{1}{t_s}(u_{b,d} - u_{b,g})$$

$$\frac{dv_{b,d}}{dt} + \frac{\kappa^2}{2\Omega}u_{b,d} = -\frac{1}{t_s}(v_{b,d} - v_{b,g})$$

Lazy way to model local warps

We look for equilibrium solutions oscillating with Keplerian frequency

$$u_{b,g} = \text{Re} \left[\psi \Omega t_s \left(\frac{i \Omega t_s (\beta + \gamma) + f \beta}{\beta (\beta + \gamma (1 + f))} \right) e^{i \Omega t} \right]$$

$$v_{b,g} = \text{Re} \left[-\frac{1}{2} \psi \kappa^2 t_s^2 \left(\frac{\beta + \gamma}{\beta (\beta + \gamma (1 + f))} \right) e^{i \Omega t} \right]$$

$$u_{b,d} = \text{Re} \left[\psi \Omega t_s \left(\frac{i \Omega t_s \gamma - \beta}{\beta (\beta + \gamma (1 + f))} \right) e^{i \Omega t} \right]$$

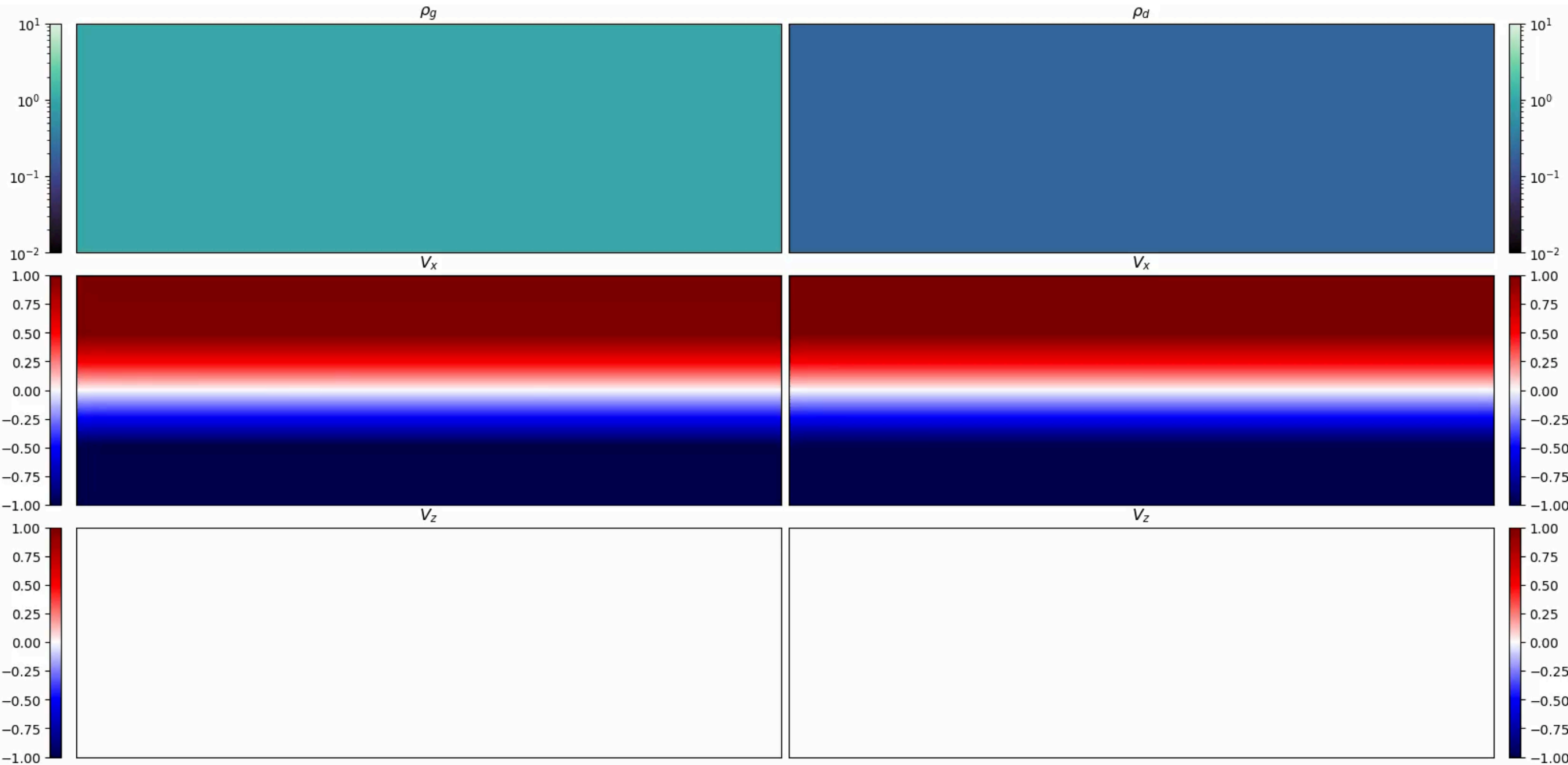
Initial Conditions

$$v_{b,d} = \text{Re} \left[-\frac{1}{2} \psi \kappa^2 t_s^2 \left(\frac{\gamma}{\beta (\beta + \gamma (1 + f))} \right) e^{i \Omega t} \right]$$

$$\beta = t_s^2 (\kappa^2 - \Omega^2) \quad \gamma = 2i \Omega t_s + 1 + f$$

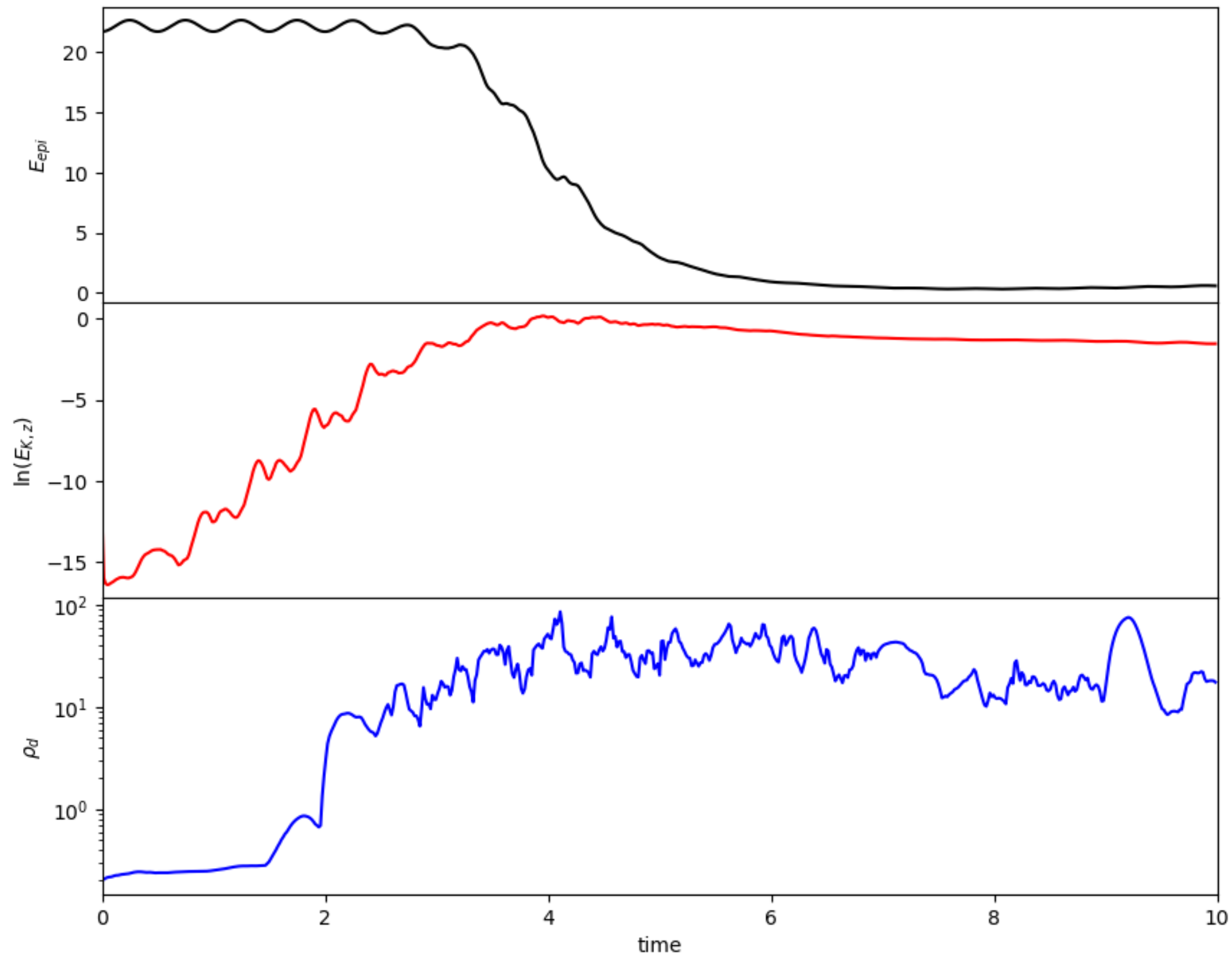
Results (code: Idefix)

Fiducial Simulation: $\psi = 0.1$ $\kappa - \Omega = 0.01$ $\text{St} = 1$ $f = 0.2$



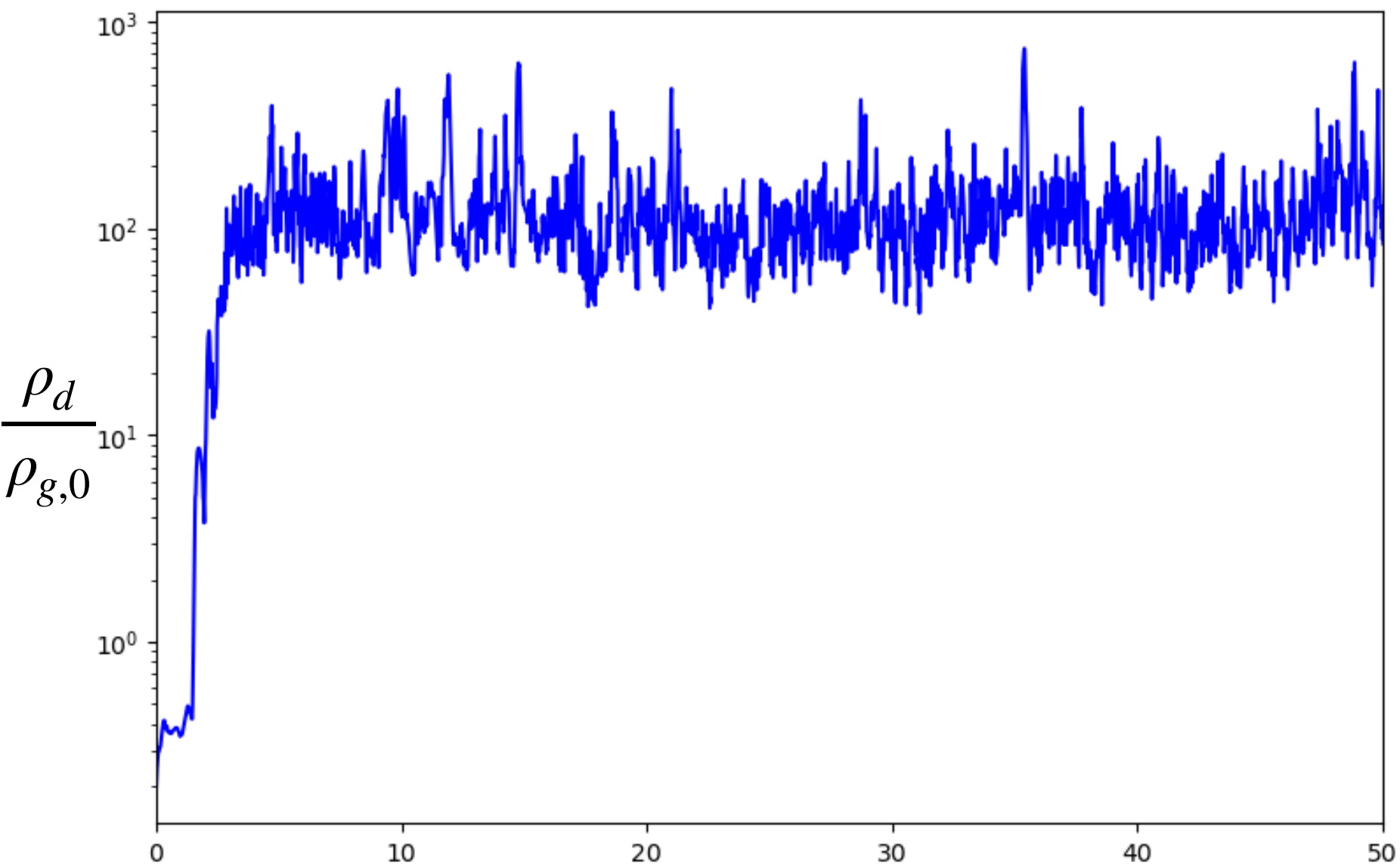
Results (code: ldefix)

Fiducial Simulation: $\psi = 0.1$ $\kappa - \Omega = 0.01$ $St = 1$ $f = 0.2$



Results (code: Idefix)

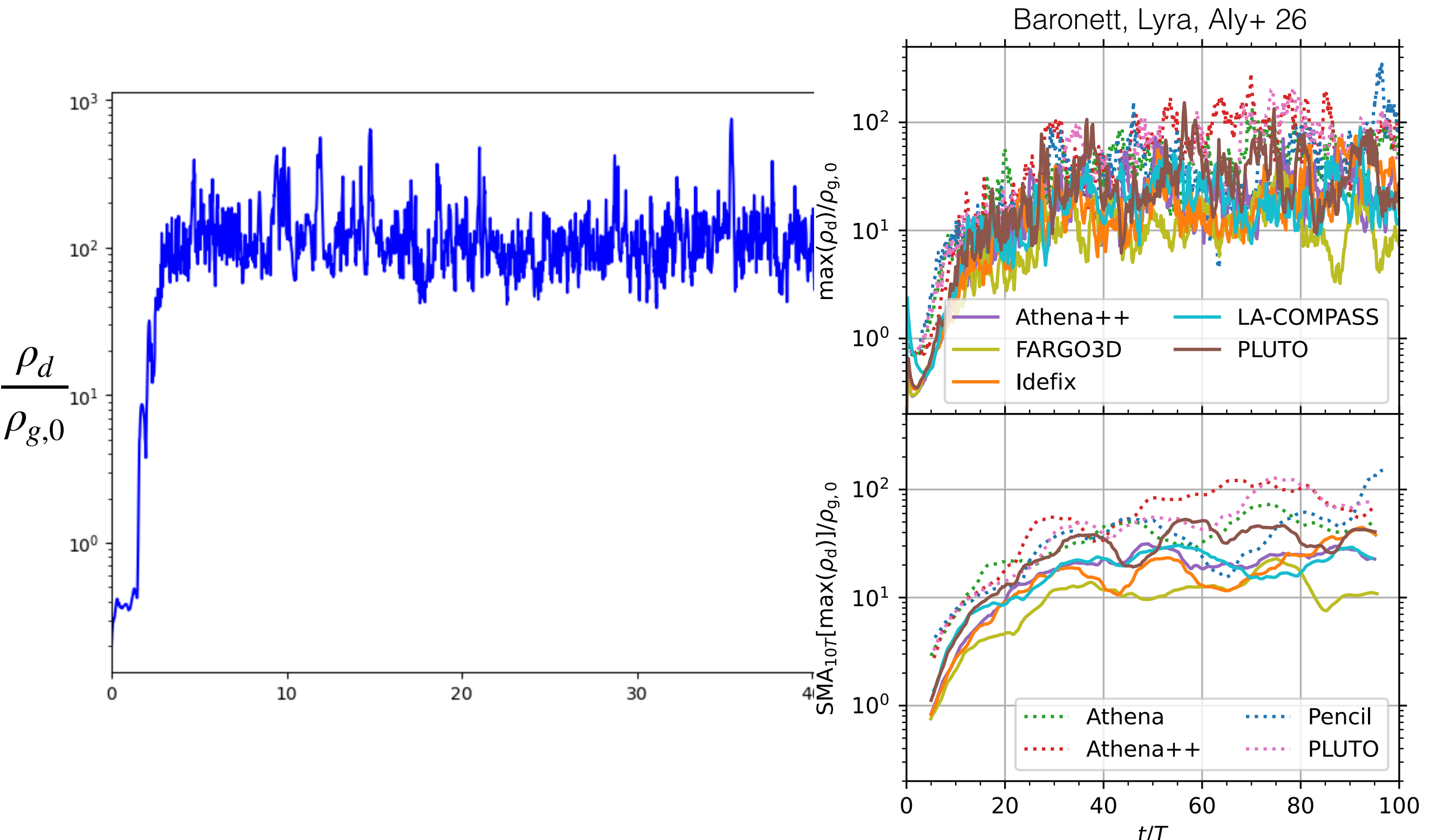
Fiducial Simulation: $\psi = 0.1$ $\kappa - \Omega = 0.01$ $St = 1$ $f = 0.2$



Feedback doesn't matter!!!

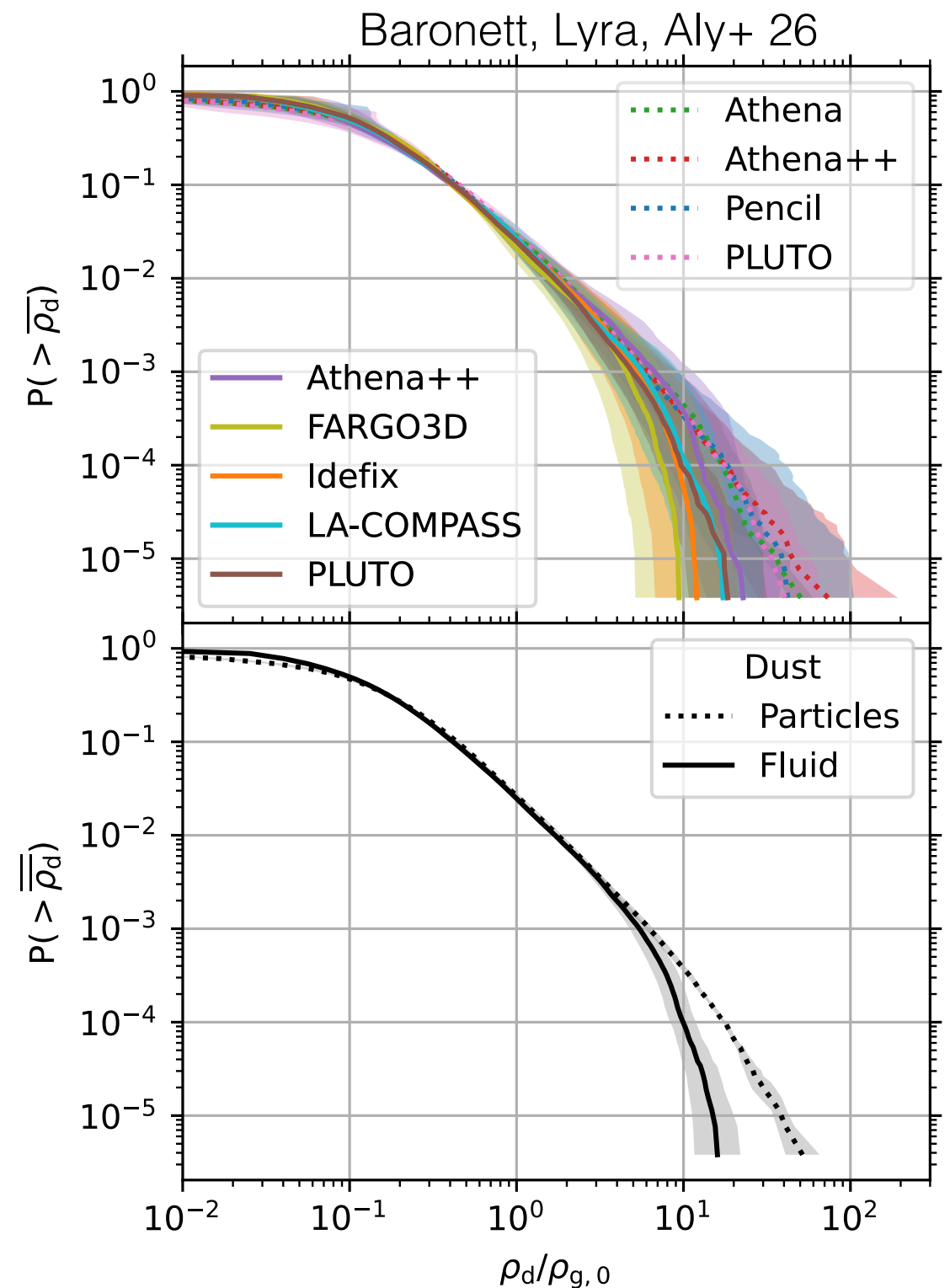
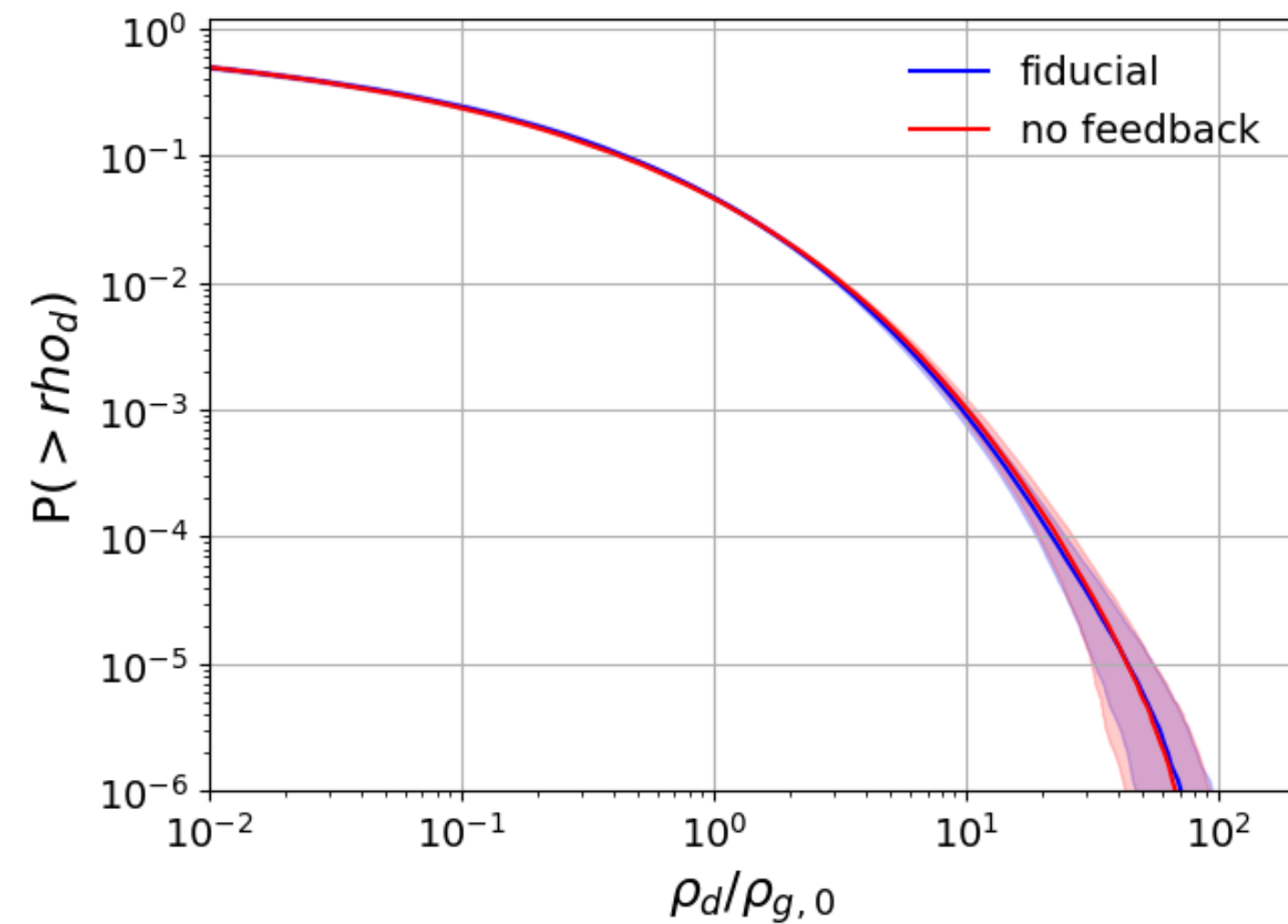
Dusty warps vs SI

Fiducial Simulation: $\psi = 0.1$ $\kappa - \Omega = 0.01$ $St = 1$ $f = 0.2$

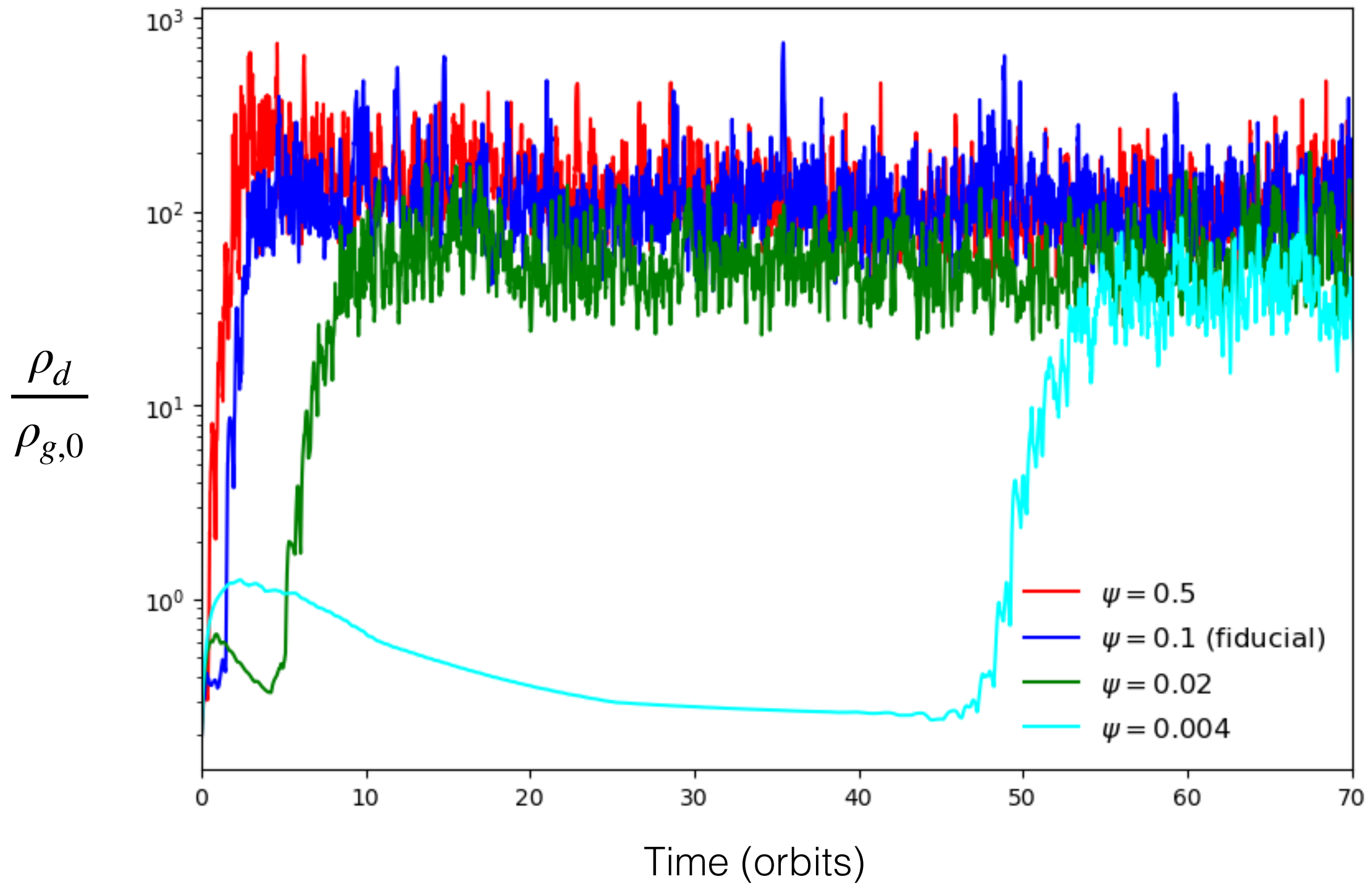


Dusty warps vs SI

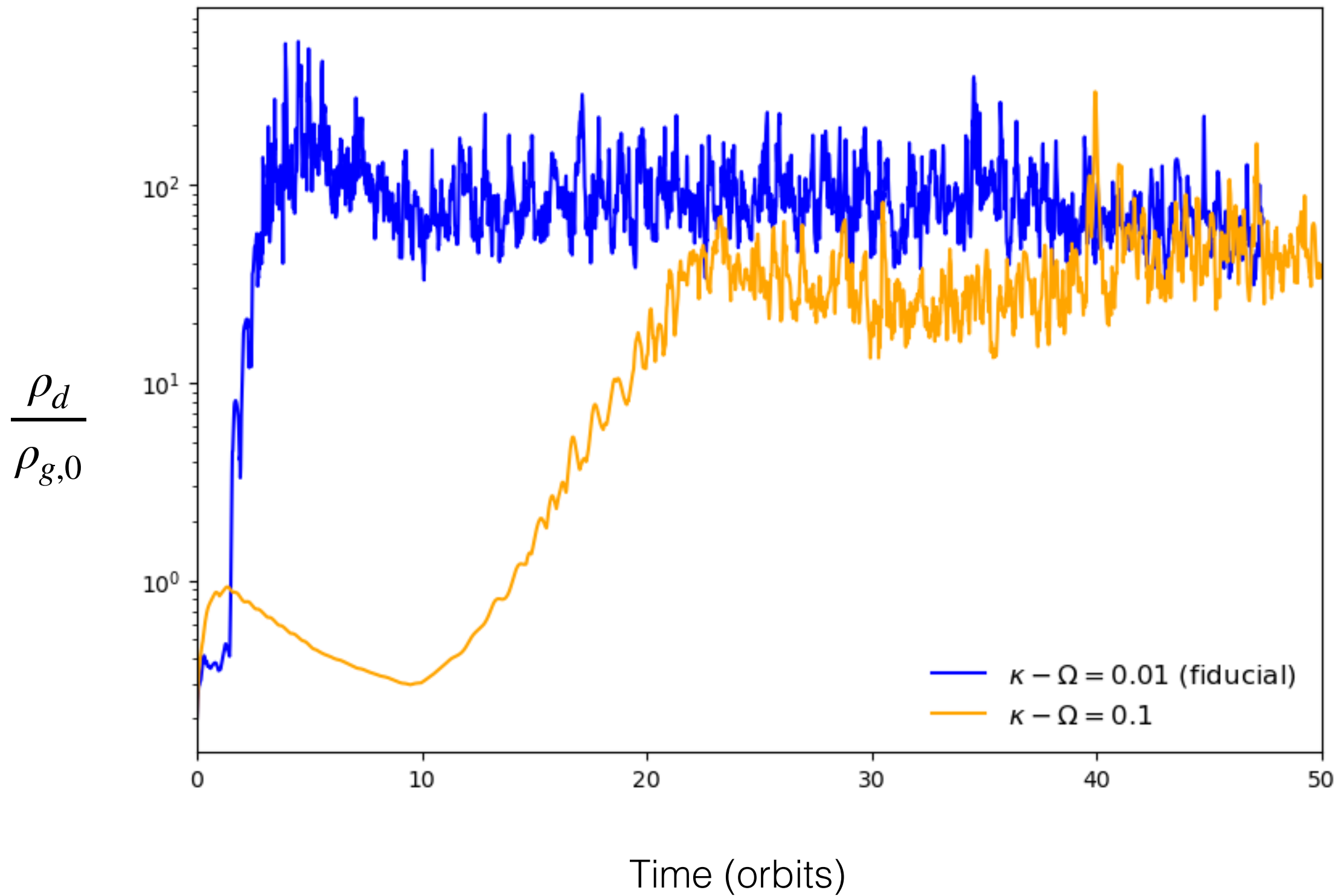
Fiducial Simulation: $\psi = 0.1$ $\kappa - \Omega = 0.01$ $St = 1$ $f = 0.2$



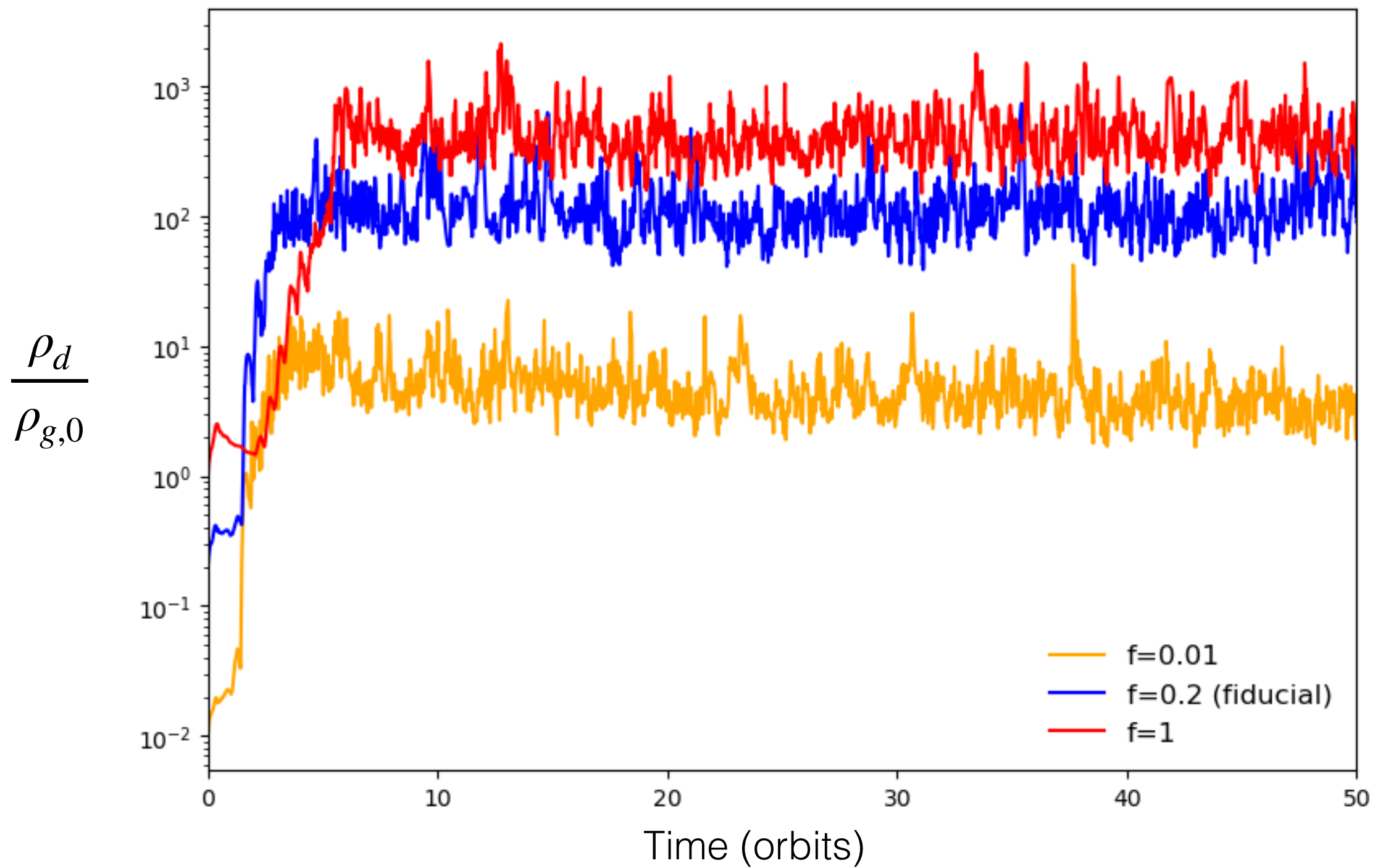
Effect of Warp Parameters



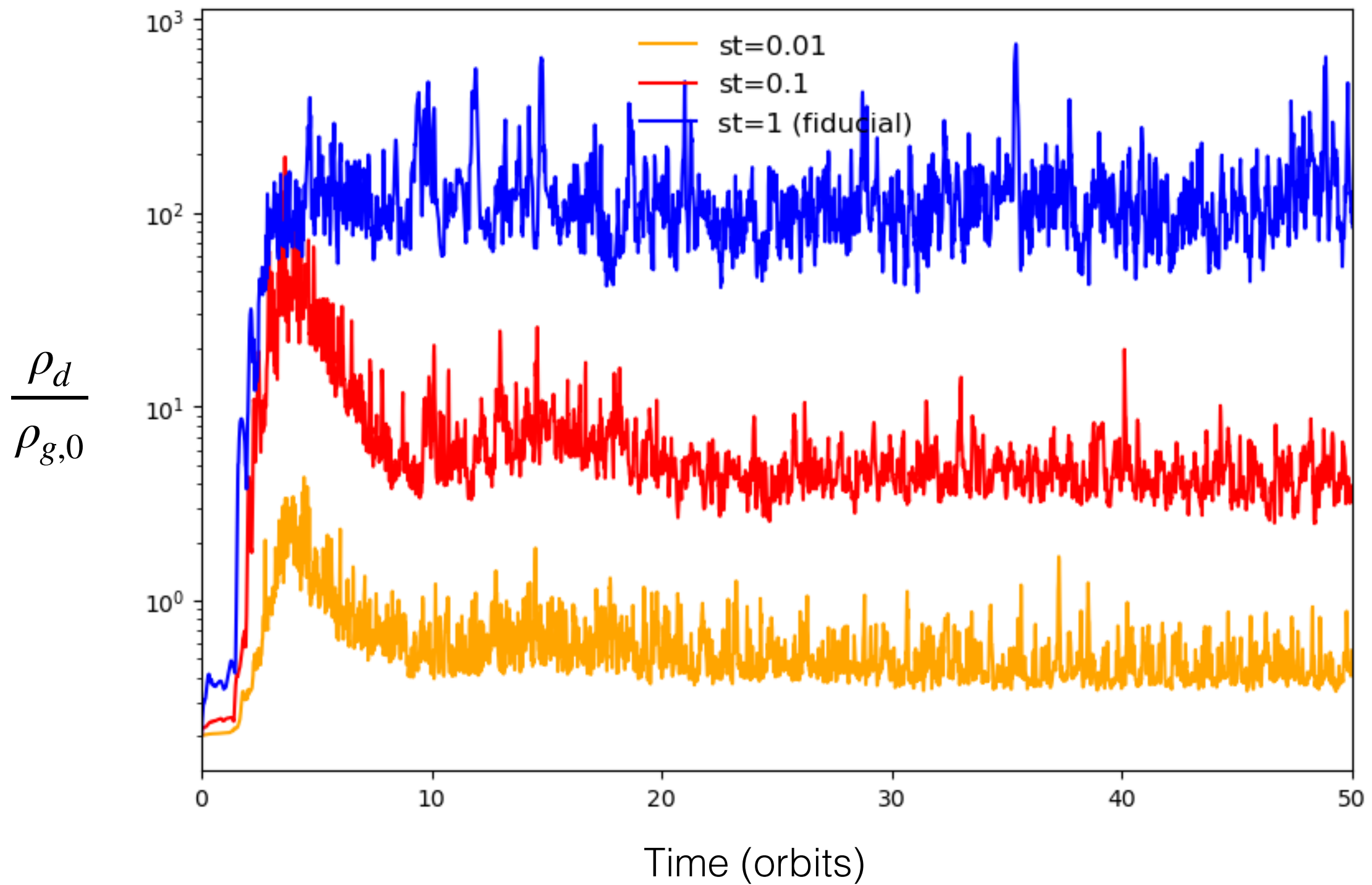
Effect of Warp Parameters



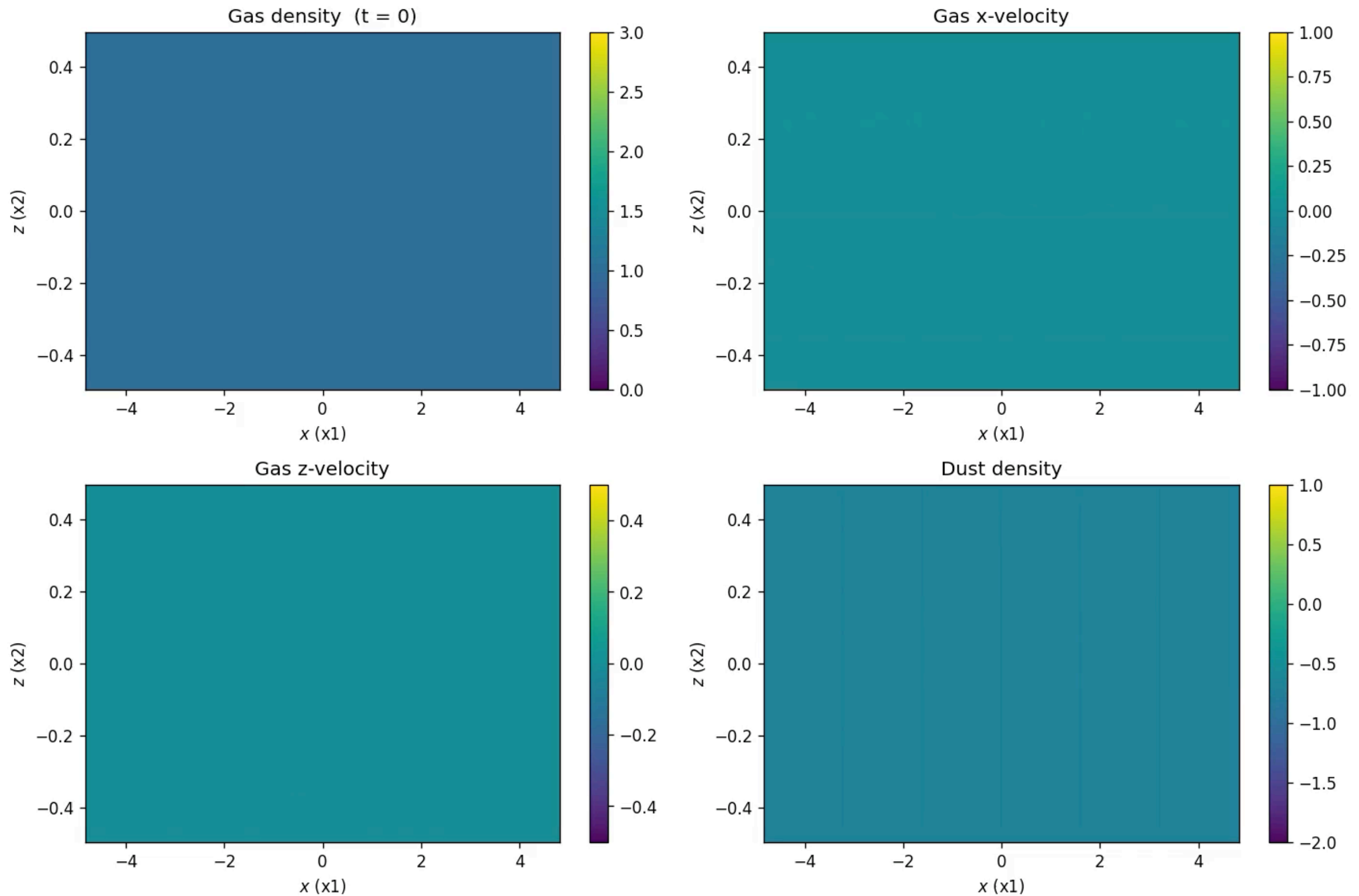
Effect of Dust Parameters



Effect of Dust Parameters



Verification with Athena++ (dust as particles) Callum's runs



Summary

- Local instability, potentially related to global WInDI
- Much faster and achieves higher density than SI
- Is not affected by dust feedback
- Works for any dust to gas ratio, but requires $St \sim 1$
- Results are robust for different BC, codes, dust models
- Theory predicts dust quenches PI, strongest at $St \sim 1$

Parametric Instability with dust: free sloshing Callum's work

- Include dust, asymptotic expansion in ψ , multiple timescale analysis
- Growth rates:

